MATH808K - Brauer Groups

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1 Overview

The goals of the this course:

1. Define central simple k-algebras(CSA/k)
2. Classify CSA's
3. Define the Brauer group(motivate Galois cohomology)
4. Blackbox Galois cohomology
5. Computing Brauer gorups of some special fields

Example 1.1 (CSA/k(Central simple k-algebras)).

1. $\mathcal{C}$ is a category, $\mathrm{Hom}_{\mathcal{C}}(A,B)$ are finite dim vector spaces, $\mathrm{End}_{\mathcal{C}}(A)$ are finite dim k-algebras
2. $\mathcal{C}'$ is the category of finite dim vector spaces, $\mathrm{End}_{\mathcal{C}'}(k^{\oplus n}) = M_n(k)$
3. $\mathcal{C}$ is the cat of local Artinian k-algebras, $\mathrm{Hom}_{\mathcal{C}}(A,B) \subseteq \mathrm{Hom}_{\mathcal{C}'}(A,B)$
4. X proj k-var and V, W vector bundles over X

$$\mathrm{Hom}\ (V,W) = \Gamma(X, \mathrm{Hom}\ (V,W)) = \Gamma(X, V^\vee \otimes W) = H^0(X, V^\vee \otimes W)$$

Example 1.2 (Examples of finite k-algebras). F/k finite field extension, $k[x]/(x^2)$, $k[x,y]/(x^2,y^2,xy)$, $M_n(k)$, $M_n(F)$

2 Quaternion Algebras

First let's assume Char $k \neq 2$

Definition 2.1. $a, b \in k^\times$, the *generalized quaternion algebra* is denoted as $(a,b) = \underset{2}{k}[i,j]/(i^2 = a, j^2 = b, ij = -ji)$, for $q = x + yi + zj + wij$, write $\bar{q} = x - yi - zj - wij$, and $\| q \| = q\bar{q} = \bar{q}q = x^2 + ay^2 + bz^2 + abw^2$

Definition 2.2. We say a k-algebra A has *division* if A is a division ring. We say A *split* if $A \cong M_n(k)$ for some n. We call $f : A \xrightarrow{\cong} M_n(k)$ a splitting of A

Example 2.3. $(-1,-1)/\mathbb{R}$ has divsion: $q^{-1} = \frac{\bar{q}}{\|q\|^2} = \frac{x-yi-zj-wij}{x^2+y^2+z^2+w^2}$ for $q = x + yi + zj + wij$.
$(-1,-1)/\mathbb{C}$ splits with splitting $f : Q \to M_2(\mathbb{C})$, $f(i) = \begin{bmatrix} -i & \\ & i \end{bmatrix}$, $f(j) = \begin{bmatrix} & -1 \\ 1 & \end{bmatrix}$

Proposition 2.4. Suppose $Q = (a,b)/k$, then the following are equivalent

1. Q split
2. Q doesn't have division
3. There is a nontrivial solution over k to $x^2 - ay^2 - bz^2 + abw^2 = 0$
4. There is a solution over k to either $a = x^2 - by^2$ or $b = z^2 - aw^2$

Remark. $(-1,-1)/\bar{k}$ always split
Proof.

- (1)⇒(2): E_{ij} has no inverse
- (2)⇒(3): Assume not, then $q^{-1} = \frac{\bar{q}}{\|q\|^2}$ is defined
- (3)⇒(4):
- (4)⇒(1):

□

Lemma 2.5.

1. $(a,b) \cong (b,a)$
2. $(a,b) \cong (au^2,b)$ for all $u \in k^\times$
3. $(1,b) \cong M_2(k)$

Proof.

1. Switch i,j
2. Consider $i \mapsto ui$
3. $i \mapsto \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}, j \mapsto \begin{bmatrix} 0 & b \\ 1 & 0 \end{bmatrix}$

□

Proposition 2.6 (Classification of finite division k-algebra of dimension 1,2,3,4).

1. $\dim_k A = 1$: $A = k$
2. $\dim_k A = 2$: $A = k1 \oplus ki$, hence A is commutative, therefore A/k is a quadratic field extension
3. $\dim_k A = 3$: Consider $Z(A)$, we have the following cases
   1. $\dim_k Z(A) = 3$: A is commutative, hence A/k is a cubic field extension
   2. $\dim_k Z(A) = 2$: we may assume $A = k1 \oplus ki \oplus kj$ and $Z(A) = k1 \oplus ki$, but then $ij = ji$ and $\dim_k Z(A) = 3$
   3. $\dim_k Z(A) = 1$: we may assume $A = k1 \oplus ki \oplus kj$, $Z(A) = k$ and $ij \neq ji$. Let's write $i^2 = a + bi + cj$, then $ai + bi^2 + cij = ii^2 = i^2 i = ai + bi^2 + cji \Rightarrow c(ij - ji) = 0 \Rightarrow c = 0$, hence $i^2 = a + bi$. Then multiply by i as a matrix would look like $\begin{bmatrix} 0 & a & x \\ 1 & b & y \\ 0 & 0 & z \end{bmatrix} =: M$ for some $x,y,z \in k$, so $\exists v \neq 0$ such that $iv = zv$, but then $(i-z)v = 0$ which is impossible
4. $\dim_k A = 4$: $A \cong (a,b)$ for some $a,b \in k^\times$

Lemma 2.7. Let D be a 4 dimensional, k-central division algebra, and assume there exists a k-subalgebra $E \subseteq A$ so that $E \cong k(\sqrt{a})$, $a \notin k^{\times^2}$, then $A \cong (a,b)$ for some $b \in k^\times$

If Char $k = 2$(note that $-1 = 1$), things are more complicated

Definition 2.8. $a \in k, b \in k^\times$, define $[a,b) = k[i,j]/(i^2 + i = a, j^2 = b, ij = ji + j)$, suppose $q = x + yi + zj + wij$, then $\bar{q} = x + y(1+i) + zj + wij = (x+y) + yi + zj + wij$, since the minimal polynomials of i, j, ij are $x^2 + x + a$, $x^2 + b$ and $x^2 + ab$ [since $(ij)^2 = ijij = i(ij + j)j = i^2 j^2 + ij^2 = (i+a)b + ib = ab$]

Exercise 2.9. Suppose Char $k = 2$, the following are equivalent

1. $[a,b)$ split
2. $[a,b)$ doesn't have division
3. b is a norm from $k(\alpha)/k$ where α is a root of $x^2 + x + a = 0$
4. The conic $ax^2 + by^2 = z^2 + zw$ has a k point

3 Projective Conics

For $a,b \in k^\times$, consider

$$\begin{bmatrix} y_0 \\ y_1 \\ y_2 \end{bmatrix} = \begin{bmatrix} 0 & b & 0 \\ a & 0 & 0 \\ 0 & 0 & ab \end{bmatrix} = \begin{bmatrix} x_0 \\ x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} bx_1 \\ ax_0 \\ abx_2 \end{bmatrix}$$

Then conics $V(ay_0^2 + by_1^2 - y_2^2) = V(a(bx_1)^2 + b(ax_0)^2 - (abx_2)^2) = V(ax_0^2 + bx_1^2 - abx_2^2)$

Proposition 3.1. $(a,b),(c,d)$ are two quaternion algebras over k, then $(a,b)\cong(c,d)$ iff $C_{a,b}=V(ax_0^2+bx_1^2-x_2^2)$ and $C_{c,d}=V(cx_0^2+dx_1^2-x_2^2)$ define the same conics up to $PGL_2(k)$

Proposition 3.2. $(a,b)/k$ split iff $C_{a,b}(k)\neq\varnothing$

Let's briefly recall Hensel's lemma

Lemma 3.3 (Hensel's lemma). $f(x)\in\mathbb{Z}_p[x]$, $a\in\mathbb{Z}_p$, $f(a)\equiv 0 \bmod p$, $f'(a)\not\equiv 0 \bmod p$, then there exists $\alpha\in\mathbb{Z}_p$ such that $f(\alpha)=0$ in $\mathbb{Z}_p$ and $\alpha\equiv a \bmod p$

Example 3.4 (Non-isomorphic quaternion algebras with divisions). $Q_1=(-1,7)/\mathbb{Q}\ncong Q_2=(-1,3)/\mathbb{Q}$.

- Check Q_1 has division, suppose not, then $7=z^2+w^2$ has a solution, but since $0=z^2+w^2$ has no non-trivial solution in $\mathbb{F}_7$, hence we have $7=(7\alpha)^2+(7\gamma)^2=49(\alpha^2+\gamma^2)$ which is impossible
- Check Q_2 has division, suppose not, then $3=z^2+w^2$ has a solution, but since $0=z^2+w^2$ has no non-trivial solution in $\mathbb{F}_3$, hence we have $3=(3\alpha)^2+(3\gamma)^2=9(\alpha^2+\gamma^2)$ which is impossible
- $(-1,3)/\mathbb{Q}_7$ split, consider $x^2+y^2-3z^2-3w^2=0$, let $y=z=1$, $w=0$ and $f(x)=x^2-2$, then $f(3)\equiv 0 \bmod 7$ and $f'(3)\equiv 1 \bmod 7$, by Hensel's lemma 3.3 we know there exists $\mathbb{Q}_7\ni\alpha\equiv 3$ such that $f(\alpha)=0$

Example 3.5. Quaternion algebras over $\mathbb{F}_q(q=p^n,p\neq 2)$ split, just consider $1^2-ax^2-by^2+0^2=0\Rightarrow ax^2=1-by^2$, write $S_a=\{ax^2\}$, $S_b=\{1-by^2\}$, then $\mid S_a\mid=\mid S_b\mid=\mid\{x^2\}\mid=\frac{q-1}{2}+1=\frac{q+1}{2}$, so S_a and S_b cannot be disjoint

4 Central Simple Algebras

Definition 4.1. A k-algebra A is a *simple* if the only two-sided ideals of A is 0 and A. A *$k$central algebra* is a finite k-algebra with $Z(A)=k$

Example 4.2. $M_n(k)$ as a left module is not simple, for example, consider left ideal generated by $\begin{pmatrix}1&0\\1&0\end{pmatrix}$

Example 4.3.

1. $M_n(k)$
2. Quaternion algebra A/k, F/k is a quadratic field extension such that A splits over F, i.e. $A\otimes_k F\cong M_2(F)$, thus $Z(A)\otimes_k F\subseteq Z(A\otimes_k F)=Z(M_2(F))=F$, which implies that $Z(A)=k$. (Similarly, if $I\subseteq A$ is a two-sided ideal, then $I\otimes_k F$ is either $M_2(F)$ or 0, which implies that $I=A$ or $I=0$ respectively)

Proposition 4.4. A,B are finite k-algebras, then $Z(A\otimes_k B)=Z(A)\otimes_k Z(B)$

Proposition 4.5. A is CSA/k, B finite k-alg, then two-sided ideals of $A\otimes_k B$ are of the form $A\otimes_k J$ where $J=I\cap(1\otimes_k B)$ is a two-sided ideal in B

Example 4.6. If $D=(a,b)/k$, $D'=(a,b')/k$, then $D\otimes_k D'\cong(a,bb')\otimes_k M_2(k)$

Example 4.7. D is a k-central division algebra, then $M_n(k)\otimes_k D\to M_n(D)$ $(a_{ij})\otimes d\mapsto(a_{ij}d)$ is an iso, in particular $M_n(k)\otimes M_m(k)\to M_{nm}(k)$ is the Kronecker product

Theorem 4.8 (Wedderburn's theorem). A is a finite simple k-alg, then $A\cong M_n(D)$ for some division k-alg and some n, D is unique up to iso, n is unique. Moreover, if A is k-central, so is D

As a corollary of a Theorem 4.8, $\dim_k(A)=n^2$, so we can define

Definition 4.9. The the *degree* of a simple k-algebra is the square root of its dimension

Definition 4.10. We say the a CSA/k A is split if $A\cong M_n(k)$ for some n, we say A is a F split, or split by F for a field extension F/k if $A\otimes_k F\cong M_n(F)$ for some n

The notion of a CSA is really irrelevant with respect to base change, as the following proposition asserts

Proposition 4.11. A is a finite k-algebra, F/k field extension. Then A is a CSA/k iff $A\otimes_k F$ is a CSA/F for all F(for some F)

Corollary 4.12. A is a CSA/k, then there is a finite field extension F/k where A splits

Proof. Let $\varphi : A \otimes_k \bar{k} \to M_n(\bar{k})$ be an isomorphism, pick a basis $\{a_i\}$ of A, let $\varphi(a_i) = \sum c_{ij} e_j$ for $c_{ij} \in \bar{k}$. Let $F = k(c_{ij})$, hten we can define $\psi : A \otimes_k F \to M_n(F)$, $\psi(a_i) = \sum c_{ij} e_j$, since $\psi \otimes_F \bar{k} = \varphi$, and since $\bar{k}$ is a faithfully flat F-module, we find ψ is an isomrophism

□

Lemma 4.13. If D is a finite division $\bar{k}$ algebra, then $D \cong \bar{k}$

Theorem 4.14. A finite k-algebra A is CSA/k iff $A \otimes_k \bar{k} \cong M_n(\bar{k})$

Proof.

□

5 Galois Splitting Fields

Definition 5.1.

6 Brauer Groups

Theorem 6.1 (Relative Skolem-Noether theorem). S is a CSA/k, $R \hookrightarrow S$ is a simple k-algebra, suppose $f, g : R \to S$ are are homomorphisms, there there exists automorphism $\alpha : S \to S$ such that $\alpha \circ f = g$

Consider A is a CSA/k with E being a maximal subfield where E/k is finite Galois with Galois group G, then for each $\sigma \in G$, we can find $x_\sigma \in A^\times$ such that $x_\sigma a x_\sigma^{-1} = \sigma(a)$, $\forall a \in E$. If y_σ is another choice, then x_σ, y_σ differ by $E^\times$, hence this defines $x_\sigma x_\tau = \alpha_{\sigma,\tau} x_{\sigma\tau}$

$\alpha : G \times G \to E^\times$ is actually a 2-cocycle

Definition 6.2. Suppose H is an abelian group with a G action, $Z^2(G, H)$ consists of $\alpha : G \times G \to H$ such that $\alpha_{\rho\sigma,\tau} \cdot \alpha_{\rho,\sigma} = \rho(\alpha_{\sigma,\tau}) \cdot \alpha_{\rho,\sigma\tau}$. $B^2(G, H)$ consists of $\gamma_{\sigma,\tau} = \frac{\sigma(f_\tau) \cdot f_\sigma}{f_{\sigma\tau}}$

Lemma 6.3. Suppose A is a finite dimensional k-algebra, $E/k \subseteq A$ is a finite Galois extension with Galois group G, and if $\{x_\sigma\}_{\sigma \in G} \subseteq A^\times$ such that $x_\sigma a x_\sigma^{-1} = \sigma(a)$, $\forall a \in E$. Then x_σ are E-linear independent in A

Proof.

□

Proposition 6.4. Suppose $\alpha : G \times G \to E^\times$ is a 2-cocycle, There is a k-algebra $(E/k, \alpha) \coloneqq \oplus E x_\sigma$ with $x_\sigma x_\tau = \alpha_{\sigma,\tau} x_{\sigma\tau}$, $x_\sigma a x_\sigma^{-1} = \sigma(a)$, $\forall a \in E$. This is a CSA/k. Moreover, $(E/k, \alpha) \cong (E/k, \beta)$ if $\alpha \sim \beta$

Proof.

□

Theorem 6.5. Consider $\phi : \mathrm{Br}\ (E/k) \to H^2(G, E^\times)$

Proof. Consider the inverse $\phi^{-1} : H^2(G, E^\times) \to \mathrm{Br}\ (E/k)$, $[\alpha] \mapsto (E/k, \alpha)$

□

Definition 6.6. Suppose H is a group with a G-action, then

$$H^0(G, H) = Z^0(G, H) = H^G = \{h \in H \mid \sigma(h) = h, \forall \sigma \in G\}$$

If H is not abelian, this is a pointed set

Remark. A sequence of pointed set is $L \xrightarrow{\pi} M \xrightarrow{\rho} N$, and every element in $\ker \rho$ has an preimage in L. If $N = \{*\}$, then π is surjective, but even if $L = \{*\}$, ρ might be non-injective

Proposition 6.7. Suppose $1 \to A \to B \to C \to 1$ is an exact sequence of groups with G-action, then there is the long exact sequence

$$1 \to H^0(G,A) \overset{(1)}{\to} H^0(G,B) \overset{(2)}{\to} H^0(G,C) \overset{\delta^1}{\to} H^1(G,A) \overset{(3)}{\to} H^1(G,B) \overset{(4)}{\to} H^1(G,C)$$

Moreover, if A is central in B, then δ^2 exists and we can extend this with $\overset{\delta^2}{\to} H^2(G,A) \to H^2(G,B) \to H^2(G,C)$

Proof.

□

Example 6.8.

$$1 \to E^\times \to \mathrm{GL}_n(E) \to \mathrm{PGL}_n(E) \to 1$$

Then we get $H^1(G,E^\times) \to H^1(G,\mathrm{PGL}_n(E))(\cong CSA/k) \overset{\delta^2}{\to} H^2(G,\times E)(\cong \mathrm{Br}\ (E/k))$

Proposition 6.9.

1. For CSA/k A,B with $\deg A = n$, $\deg B = m$, $\delta^2_{nm}([A \otimes_k B]) = \delta^2_n([A])\delta^2_m([B])$
2. For CSA/k A with $\deg A = n$, $\delta^2_n([A]) = 0 \Leftrightarrow A \cong M_n(k)$
3. If $[E:k] = n$, then δ^2_n is surjective, and $\phi^{-1} \circ \delta^2_n([(E/k,\alpha)]) = [(E/k,\alpha)^{\mathrm{op}}]$

Proposition 6.10. Suppose A is a CSA/k with ind $A = d$, then ind $A^{\otimes r} \mid \binom{d}{r}$, $r \le d$. In particular, $A^{\otimes r}$ split, and Br (E/k) are torsion

Proof. Let E/k be finite Galois extension that splits A, $G = \mathrm{Gal}\ (E/k)$, $V = E^{\oplus d}$, we get the following commutative diagram

$$\begin{array}{ccccccccc}
1 & \longrightarrow & E^\times & \longrightarrow & \mathrm{GL}\ (V) & \longrightarrow & \mathrm{GL}\ (V) & \longrightarrow & 1 \\
 & & \lambda \mapsto \lambda^r \downarrow & & \pi \downarrow & & \downarrow & & \\
1 & \longrightarrow & E^\times & \longrightarrow & \mathrm{GL}\ (\bigwedge^r V) & \longrightarrow & \mathrm{PGL}\ (\bigwedge^r V) & \longrightarrow & 1
\end{array}$$

Here $\pi(\phi) = \bigwedge^r \phi$, so we get exact squares

□

7 Group Cohomology Spectral Sequence

Definition 7.1. Let $A \cong M_n(D_A)$ be a CSA/k with underlying division algebra D_A. The *index of* A is defined to be ind $(A) = \deg(D_A)$. The index depends only on the class of A in Br (k)

Lemma 7.2 (Shapiro's lemma). B is a $\mathbb{Z}H$-module, then $H^i(G, M_H^G(B)) \cong H^i(H,B)$

Proof.

□

Definition 7.3. Suppose $[G:H] < \infty$ and A is a $\mathbb{Z}G$-module, S is a set of representatives of cosets of G/H, for any $\varphi \in M_H^G(A) = \mathrm{Hom}\ (\mathbb{Z}G, A)$ we can define

$$\varphi_H^G : \mathbb{Z}G \to A, x \mapsto \sum_{s \in S} s\varphi(s^{-1}x)$$

It is easy to check that φ_H^G is defined independently from S. So we get a map $M_H^G(A) \to A$ which induce the *corestriction map* $\mathrm{cor}_G^H : H^i(H,A) \to H^i(G,A)$

Proposition 7.4. Suppose $K \leq H \leq G$

1. $\mathrm{res}^G_K = \mathrm{res}^H_K \circ \mathrm{res}^G_H$
2. If $[G : K] < \infty$, then $\mathrm{cor}^K_G = \mathrm{res}^K_H \circ \mathrm{res}^H_G$
3. If $[G : H] < \infty$, then $\mathrm{cor}^H_G \circ \mathrm{res}^G_H$ is multiplication by $[G : H]$

Proof.
1.
2.
3.

□

8 Reduced Norm

$\mathrm{Br}\ (\bar{k}) = 0$, $\mathrm{Br}\ (\mathbb{F}_q) = 0$, $\mathrm{Br}\ (\mathbb{R}) = \mathbb{Z}/2\mathbb{Z}$ is generated by $(-1, -1)$, note that $\mathrm{Br}\ (\mathbb{R}) = \mathrm{Br}\ (\mathbb{C}/\mathbb{R}) = \mathbb{R}^\times / N_{\mathbb{C}/\mathbb{R}}(\mathbb{C}^\times) = \mathbb{R}^\times / \mathbb{R}_{>0}$, however $\mathrm{Br}\ (\mathbb{Q}) \neq 0$ and $\mathrm{Br}\ (\mathbb{Q}_p) \neq 0$

Definition 8.1. A/k is a CSA, pick E/k finite Galois splitting field for A with splitting $f : A \otimes_k E \to M_n(E)$, then the following map

$$A \cong A \otimes_k k \subseteq A \otimes_k E \underset{\cong}{\overset{f}{\to}} M_n(E) \overset{\det}{\to} E$$

is called the *reduced norm* Nrd of A

Remark. $\mathrm{Nrd} : A \to E$ is independent of the splitting, suppose $f, g : A \otimes_k E \overset{\cong}{\to} M_n(E)$ are splittings, then we have the commutative diagram

$$\begin{array}{ccccccc}
A & \hookrightarrow & A \otimes_k E & \underset{\cong}{\overset{f}{\longrightarrow}} & M_n(E) & \xrightarrow{\det} & E \\
\| & & \| & & \downarrow{\scriptstyle g \circ f^{-1}} & & \| \\
A & \hookrightarrow & A \otimes_k E & \underset{\cong}{\overset{g}{\longrightarrow}} & M_n(E) & \xrightarrow{\det} & E
\end{array}$$

Nrd is independent of the choice of the splitting field E, for some reasoning about composite fields

Remark. $\mathrm{Nrd} : A \to k$ is multiplicative, i.e. $\mathrm{Nrd}\ (xy) = \mathrm{Nrd}\ (x)\ \mathrm{Nrd}\ (y)$, so $\mathrm{Nrd} : A^\times \to k^\times$ is a group homomorphism

Proposition 8.2. A/k is a CSA, $\deg A = n$, $N_{A/k} : A \to k$, $a \mapsto \det m_a$ is the *field norm* map, $N_{A/k} = \mathrm{Nrd}^n$

Proof. Note that $N_{A/k}(a) = N_{A \otimes_k E/E}(a \otimes 1)$ and that $\mathrm{Nrd}\ (a) = \mathrm{Nrd}\ (a \otimes 1)$. So we may assume that $E = k$, if we write elementary matrices E_{ij} as a basis of $M_n(k)$, so a matrix $M \in M_n(k)$ as an linear operator on $M_n(k)$ as the form

$$A_M = \begin{bmatrix} M & & \\ & \ddots & \\ & & M \end{bmatrix}$$

Hence $N_{A/k}(M) = \det(A_M) = (\det M)^n = \mathrm{Nrd}\ (M)^n$

□

Proposition 8.3. Nrd is a homogeneous polynomial in n^2 variables(a basis of A over k) of degree n with k-coefficients

Proof. $a = \sum_{k=1}^{n^2} \lambda_k v_k$, so $\mathrm{Nrd}\ (a) = \det(f(a \otimes 1))$

□

Definition 8.4. k is a quasi algebraically closed (or C_1) field if any homogeneous polynomial $f \in k[x_1, \cdots, x_n]$ with $\deg f = d < n$, there is some $a_1, \cdots, a_n$ not all zero with $f(a_1, \cdots, a_n) = 0$

Proposition 8.5. Algebraically closed fields are C_1

Proof. Write $f \in k[x_1, \cdots, x_n]$ as $\phi_m x_1^m + \cdots + \phi_0$ and then do induction on the variables

□

Lemma 8.6. L/k finite extension, k is C_1, then so is L

Proof. Suppose $\{v_j\}_{j=1}^m$ is a k-basis of L, $f \in L[x_1, \cdots, x_n]$ is a degree $d < n$ homogeneous polynomial, write $x_i = \sum_{j=1}^m x_{ij} v_j$, Then $f(x_1, \cdots, x_n) = g(x_{11}, \cdots, x_{nm})$, note that products of v_j's can written as the k-linear combinations of v_j's, g can be seen as a $k[x_{11}, \cdots, x_{nm}]$-linear combination of v_j's, view this as an linear operator on the k vector space L, denote its determinant h, which is then a homogeneous polynomial of degree md with coefficients in k, since $md < mn$, h has non trivial solution, which says that f also has a non-trivial solution.

□

Proposition 8.7. If k is C_1, then $\mathrm{Br}\ (k) = 0$

Proof. If k is finite, we already know that $\mathrm{Br}\ (k) = 0$. We can actually show that finite fields are C_1
Assume k is infinite, A/k is a central division algebra over k of degree $n \geq 1$, then by Theorem 8.3 we know that $\mathrm{Nrd} : A \to k$ is a homogeneous polynomial of degree n in n^2 variables with k-coefficients. If $n > 1$, then $\mathrm{Nrd}\ (x) = 0$ for some $0 \neq x \in A$ which is impossible, so $n = 1$, and $\mathrm{Br}\ (k) = 0$

□

Theorem 8.8 (Tsen's theorem). k is algebraically closed, C is a curve over k, then its function field $F = k(C)$ is C_1

Remark. This is equivalent of saying that F is a finite extension of $k(t)$, e.g. $F = \mathbb{C}(t)$, $F = \mathbb{C}(t)(\sqrt{t(t-1)(t+1)})$

Proof. By clearing out the denominators, we may assume f is a degree d homogeneous polynomial in $k[t][x_1, \cdots, x_n]$. Suppose r is the maximal degree of the coefficients in f. If we write for some sufficiently large N

$$x_i = \sum_{j=0}^{N} a_{ij} t^j$$

Then we can rewrite f as

$$\sum_{l=0}^{dN+r} f_l(a_{10}, \cdots, a_{nN}) t^l$$

Then f_l are homogeneous polynomial of $a_{10}, \cdots, a_{nN}$ of degree d, so $V(f_0, \cdots, f_{dN+r})$ is a closed subvariety of $\mathbb{P}^{n(N+1)-1}$, then we can apply Lemma 8.9, there exists $a_{10}, \cdots, a_{nN}$ not all zero such that $f = 0$, so x_i would be a solution.

□

Lemma 8.9. $W \subseteq \mathbb{P}^n$ is a positive dimensional closed subvariety, $f \neq 0$ is a rational function on $\mathbb{P}^n$, then $W \cap V(f) \neq \varnothing$

Proof. This is equivalent to $W \subseteq D(f)$, we know that $D(f)$ is affine so we could assume $D(f) = \mathrm{Spec}\ A$, then a morphism $W \hookrightarrow D(f)$ corresponds to $A \to \Gamma(W, \mathcal{O}_W) = k$ since W is proper and projective

□

Remark. If $k = \overline{\mathbb{F}_p}$, $A/\mathbb{F}_q(C)$ is a CSA, then A split over $k(C)$

9 Unramified Splitting Field For Complete Discretely Valued Fields

Definition 9.1. A *discrete valuation* on a field F is $v : F \to \mathbb{Z} \cup \{\infty\}$ satisfying

1. $v(x) = \infty \Leftrightarrow x = 0$
2. $v(x+y) \geq \min(v(x), v(y))$, equality holds iff $v(x) \neq v(y)$
3. $v(xy) = v(x) + v(y)$

A valuation is when replacing $\mathbb{Z}$ by some totally ordered abelian group(called the value group). v is trivial if $v(F^\times) = 0$

Definition 9.2. A *discrete valuation ring* is an integral domain R with a discretely valued fraction field $F = \text{Frac } (R)$ such that $R = \{x \in F \mid v(x) \geq 0\}$. In this case $m = \{x \in F \mid v(x) > 0\}$ is the unique prime ideal in R, and $m = (\pi)$, we call π the *uniformizer*, and for $x \in F$, $v(x)$ is the maximal power of π in a.

Definition 9.3. Pick some $0 < c < 1$, we can define norm $|\cdot| : F \to \mathbb{R}_{\geq 0}$ as $\mid x \mid = c^{v(x)}$, this defines a metric topology(which can be shown to be independent of the choice of c)

Example 9.4. $\mathbb{Z}_{(p)} \subseteq \mathbb{Q}$ with the p-adic valuation, $c = p^{-1}$

Definition 9.5. By completing the metric topology, we obtain a complete discretely valued field $\hat{F}$ with extended norm $|\cdot|$, and the closure of R in F is denoted by $\hat{R}$ which can be proven is $\hat{R} = \lim_{\leftarrow} R/m^n$

Example 9.6. $k[t]_{(t)} \subseteq k(t)$ with t-adic valuation, $\widehat{k(t)} = k((t))$ is the field of formal Laurent series, $\widehat{k[t]_{(t)}} = k[[t]]$ is the ring of formal power series

Definition 9.7. Consider finite separable extension E/F, then $x \in E$, $\mid x \mid_E = \mid N_{E/F}(x) \mid_F^{\frac{1}{[E:F]}}$. If F was complete wrt $|\cdot|_F$, then E is complete wrt $|\cdot|_E$ and has a valuation $w : E^\times \to \mathbb{Z}$ satisfying

1. The valuation ring of w is S the integral closure of R in E
2. $\exists e$, such that $\forall x \in R$, $w(x) = ev(x)$, and $[E : F] = ef$, where $f = [S/m' : R/m]$

Our setting: If F is complete, E/F finite and separable is said to be unramified if $f = [E : F]$ and $(S/m')/(R/m)$ is separable

Theorem 9.8. Let N_F be the category with objects unramified extensions E/F and morphisms are $E \to E'$ over F. R_F be the category of finite separable extension R/m with R/m-algebra morphism. Consider the restion functor $\text{res} : N_F \to R_F$, $E/F \rightsquigarrow (S/m')/(R/m)$, and $(E_1 \to E_2) \rightsquigarrow (S_1/m_1' \xrightarrow{f^r} S_2/m_2')$ where f^r is defined as follows: for $x \in S_1/m_1'$, let $y \in S_1$ be a lift, and let $f^r(x)$ be the image of $f(y)$ in S_2/m_2'. This functor is an equivalence of categories. So $\text{Hom}_F(E, E) = \text{Hom}_{R/m}(S/m', S/m')$ have isomorphic Galois groups

Example 9.9. $F = \mathbb{Q}_p$, E/F unramified with $[E : F] = n$, then $E = F(\xi_{p^n-1})$, and Gal (E/F) is cyclic

Now assume k is infinite, $v : k^\times \to \mathbb{Z}$ is a valuation, k is complete with respect to associated absolute values. R is the valuation ring.

Lemma 9.10. D/k is a central division algebra with $\deg D = d$, then $\exists$ valution $w : D^\times \mathbb{Z}$. And w is uniquely determined by restricting on any field $F \subseteq D$ containing k by $w(x) = v_F(x) = \frac{1}{[F:k]} v(N_{F/k}(x))$. Moreover $w(x) = \frac{1}{d} v(\text{Nrd } (x))$ and this valuation gives D the structure of a normed k-vector space

Proof. For a fixed $F \subseteq D$, $N_{D/k}(x) = \text{Nrd } (x)^d$, so D viewed as a left F-vector space is $D \cong F^{\oplus m}$ where $m = \frac{d^2}{[F:k]}$. since for $x \in F$, $\text{Nrd } (x)^d = N_{D/k}(x) = N_{F/k}(x)^m$, $mv(N_{F/k}(x)) = dv(\text{Nrd } (x)) \Rightarrow \frac{1}{[F:k]} v(N_{F/k}(x)) = \frac{1}{d} v(\text{Nrd } (x)) = w(x)$. Now let $x, y \in D^\times$ such that $x + y \in D^\times$, $F = k(x^{-1}y) \subseteq D$, we know w is a group homomorphism(since Nrd is a group homomorphism), we have

$$w(1 + x^{-1}y) = v_F(1 + x^{-}1y) \geq \min(v_F(1), v_F(x^{-1}y)) = \min(v_F(1), v_F(x^{-1}y)) \Rightarrow w(x+y) \geq \min(w(x), w(y))$$

So w is a valuation

□

Theorem 9.11. If R/m is a perfect field, then any CSA/k is split by an unramified extension of k

Proof. We will show that for any L/k unramified, any $A \in \text{CSA } /k$ is split by another unramified E/L. We prove this by induction on the index. If A/L has index one, then we can just take $E = L$. Now suppose the statement holds for all L/k unramified, $\text{ind}_L(A) \leq d$, by Lemma 9.12, A contains an unramified E/L with $[E : L] > 1$, then $\text{ind}_E(A)$ becomes less

□

Lemma 9.12. If R/m is perfect, L/k is a unramified and D is a division algebra over L with $\text{ind}_L(D) > 1$, then D contains an unramified E/L with $[E : L] > 1$

Proof. Suppose not, then for every field $L \subseteq F \subseteq D$, the residue field extension $(S/m')/(R/m)$ ($S \subseteq F$ valution ring with $m' \subseteq S$ maximal ideal) is trivial. Pick $x \in D^\times$, $\pi \in D^\times$ such that $w(\pi) = 1$. By induction we can write $y = a_0 + a_1\pi + a_2\pi^2 + \cdots + a_{n-1}\pi^{n-1} + b_n\pi^n$ which is a Cauchy sequence, so $y \in R(\pi)$

□

By Theorem 9.11, we know

$$\text{Br }(k) = \varinjlim_{E/k \text{ (finite) unramified}} \text{Br }(E/k) = \varinjlim_{E/k \text{ (finite) unramified}} H^2(\text{Gal }(E/k), E^\times)$$

And we have a split short exact sequence of $G = \text{Gal }(E/k)$-modules

$$1 \longrightarrow S^\times \longrightarrow E^\times \underset{1 \mapsto \pi}{\overset{v}{\rightleftarrows}} \mathbb{Z} \longrightarrow 0$$

So there is a split exact sequence for all $i \geq 0$

$$0 \to H^i(G, S^\times) \to H^i(G, E^\times) \to H^i(G, \mathbb{Z}) \to 0$$

Proposition 9.13. $H^i(G, S^\times) = 0$ for all $i \geq 1$

Proof. Since G is cyclic, for all $j \geq 1$, $H^{2j-1}(G, S^\times) = H^1(G, S^\times) = 1$ by Hilbert 90. $H^{2j}(G, S^\times) = H^2(G, S^\times) = (S^\times)^G/N_G(S^\times)$, here recall the norm map $N_G(x) = \prod_{g \in G} gx$. First consider commutative diagram

$$\begin{array}{ccc} S^\times & \twoheadrightarrow & (S/m')^\times \\ \downarrow{\scriptstyle N_G} & & \downarrow{\scriptstyle N_G} \\ R^\times = (S^\times)^G & \twoheadrightarrow & (R/m)^\times \end{array}$$

The right map is surjective since Gal $((S/m')(R/m))$ induce action of $(S/m')^\times$, we know $H^2(G, (S, m')^\times) = (S/m')^\times/N_G(S/m')^\times$, we want to show that the left map is surjective. For $x \in R^\times$, we can find $z_1 \in S^\times$ such that $N_G(z_1)$ has the same image as x in $(R/m)^\times$, so $\frac{x}{N_G(z_1)} = 1 + u\pi$ for some $u \in R$ Define

$$U_E^{(j)} = 1 + m'^j, \quad U_k^{(j)} = 1 + m^j$$

Consider commutative diagram

$$\begin{array}{ccc} U_E^{(j)} & \longrightarrow & S/m' \\ \Big\downarrow{\scriptstyle N_G} & & \Big\downarrow{\scriptstyle N_G} \\ U_k^{(j)} & \longrightarrow & R/m \end{array}$$

The right map(called the trace map $x \mapsto \sum_{g\in G} gx$) is surjective by lemma 9.14, so $\frac{1}{N_G(z_j)}\frac{x}{N_G(z_1\cdots z_{j-1})} = \frac{x}{N_G(z_1\cdots z_j)} = 1 + u_j\pi^j$. Note that $\left\{\prod_{i=1}^n z_i\right\}$ is Cauchy in $E^\times$, if we take z to be the limit, then $\frac{x}{N_G(z)} = 1$

□

Lemma 9.14. $H^i(G, S/m') = 0$ for all $i \geq 1$

Proof. Suppose $R/m = \mathbb{F}_q$, $S/m' = \mathbb{F}_{q^n}$ for some n. Consider $\mathbb{F}_{q^n}$ as an abelian group, since G is cyclic of order n, $\mathbb{F}_{q^n} = \frac{\mathbb{F}_q[t]}{(t^n-1)} = \mathbb{F}_q G \cong M_{\{e\}}^G(\mathbb{F}_q) \cong \mathrm{Hom}_{\mathrm{Ab}}(\mathbb{Z}G, \mathbb{F}_q)$, then apply Shapiro's lemma 7.2.

□

Lemma 9.15. The short exact sequence of $\mathbb{Z}G$ modules(as trivial modules)

$$0 \to \mathbb{Z} \to \mathbb{Q} \to \mathbb{Q}/\mathbb{Z}$$

Induces isomorphisms $H^{i-1}(G, \mathbb{Q}/\mathbb{Z}) \xrightarrow{\delta^i} H^i(G, \mathbb{Z})$

Proof. We have $H^i(G, \mathbb{Q}) \xrightarrow{\mathrm{res}} H^i(\{e\}, \mathbb{Q}) = 0 \xrightarrow{\mathrm{cor}} H^i(G, \mathbb{Q})$ for $i \geq 1$ is multiplying by $\mid G \mid$ which is an isomorphism, hence $H^i(G, \mathbb{Q}) = 0$

□

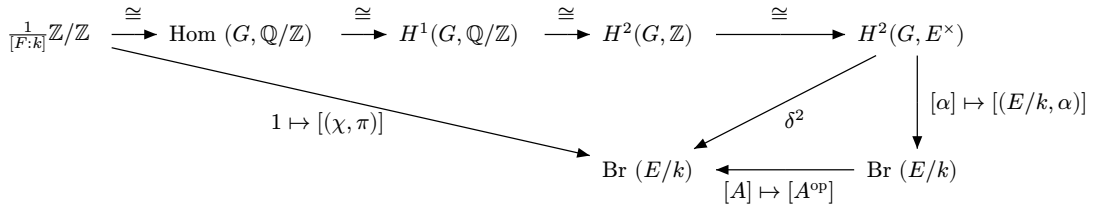
Corollary 9.16. There are isomorphisms

$$H^2(G, E^\times) \underset{\cong}{\overset{\cong}{\diamond}} H^2(G, \mathbb{Z}^\times) \underset{\cong}{\longleftarrow} H^1(G, \mathbb{Q}/\mathbb{Z})$$

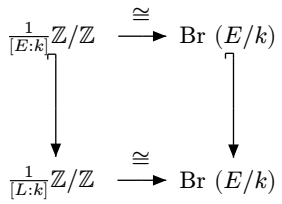
Theorem 9.17. If f is a 1-cocycle, then $f(gh) = f(g) + g(f(h)) = f(g) + f(h)$ is a group homomorphism, so as $\mathbb{Z}G$ modules, $H^1(G, \mathbb{Q}/\mathbb{Z}) = \mathrm{Hom}_{\mathrm{Ab}}(G, \mathbb{Q}/\mathbb{Z})$. Let $H = \ker f$, $F = E^H$, $\chi = \bar{f} : G/H \to \frac{1}{[F:k]}\mathbb{Z}/\mathbb{Z}$, so it can be identified with

$$\left\{(F/k, \chi), k \leq F \leq E \text{ cyclic Galois, and } \chi : \frac{\mathrm{Gal}\ (E/k)}{\mathrm{Gal}\ (F/k)} \xrightarrow{\cong} \frac{1}{[F:k]}\mathbb{Z}/\mathbb{Z} \subseteq \mathbb{Q}/\mathbb{Z}\right\}$$

And in our case, we have



If $L/E/k$ unramified, $G' = \mathrm{Gal}\ (L/k)$ and $\chi : G' \to \frac{1}{[L:k]}\mathbb{Z}/\mathbb{Z}$ such that $\overline{\chi'} : \frac{\mathrm{Gal}\ (L/k)}{\mathrm{Gal}\ (E/L)} \xrightarrow{\cong} \frac{1}{[E:k]}\mathbb{Z}/\mathbb{Z}$ is canonically χ, then



Hence Br $(k) = \cong \varinjlim \mathrm{Br}\ (E/k) \cong \mathbb{Q}/\mathbb{Z}$

Corollary 9.18. If A is a CSA/k, then $\mathrm{ind}\ (A) = \exp(A)$

Proof. Since Br $(k) = \mathbb{Q}/\mathbb{Z}$, the algebra $[A]$ generates a subgroup of size $\exp(A) = d$, so $[A]$ is the image of $\frac{1}{d} \cong H^2(\mathrm{Gal}\ (E_d/k), E_d^\times)$, where E_d is the unique unramified extension of degree d, so $d = \exp(A) \mid \mathrm{ind}\ (A) \mid d \Rightarrow \exp(A) = \mathrm{ind}\ (A) = d$

□

Corollary 9.19. Every CSA/k is a cyclic

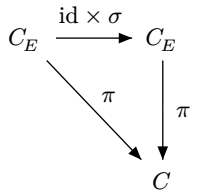
Proof. If A is a CSA/k, $\deg(A) = n$, and $\mathrm{ind}\ (A) = d$, then $d = \mathrm{ind}\ (A) \mid \deg(A) = n$, so A is split by E_d, Hence A is split by $E_d \subseteq E_n$, so $E_n \subseteq A$ is the maximal subfield of A.

□

Example 9.20. $k = \mathbb{F}_3((t))$, $E = \mathbb{F}_3((t))[x]/(x^3 - x - 1) \cong \mathbb{F}_{27}((t))$ which is actually a infinite degree extension. $A = [1, t)_{\mathbb{F}_3((t))}$ has non zero class in Br $(k) \Rightarrow$ Br $(\mathbb{F}_3((t))) \neq 0$, $A' = [1, t)_{\mathbb{F}_3(t)}$, $A' \otimes_{\mathbb{F}_3(t)} \mathbb{F}((t)) \cong A$

Setting:

- k is a field.
- C/k is a smooth geometrically connected, projective curve.
- $C_F = C \times_k F$ for any field extension F/k, denote $\pi : C_F \to C$.
- E/k is a finite Galois extension with $G = \mathrm{Gal}\ (E/k)$. $\forall \sigma \in G$



defines a G-action on C_E.

- $E(C_E) = k(C) \otimes_k E$ is the function field of C_E.
- $\mathrm{Div}\ (C_E) = \bigoplus_{x \in C_E^{(1)}} \mathbb{Z}x$ is the group of divisors of C_E.
- $\mathrm{Pic}\ (C_E)$ is the Picard group of C_E.

We have the following short exact sequence

$$1 \to E^\times \to E(C_E)^\times \overset{\mathrm{div}}{\to} \mathrm{Div}\ (C_E) \overset{D \mapsto \mathcal{O}(D)}{\to} \mathrm{Pic}\ (C_E) \to 0 \tag{9.1}$$

Lemma 9.21. There is an isomorphism of $\mathbb{Z}G$-modules

$$\mathrm{Div}\ (C_E) = \bigoplus_{x \in C^{(1)}} \left(\bigoplus_{y \in \pi^{-1}(x)} \mathbb{Z}y \right) \cong \bigoplus_{x \in C^{(1)}} M_{G_y}^G(\mathbb{Z})$$

Here G_y is the stabilizer of y

Proof. Look at the fiber over x, we may identify $\mathbb{Z}G$-modules

$$G/G_y = \{\sigma_1, \cdots, \sigma_r\} \to \pi^{-1}(x) = \{y_1, \cdots, y_r\}, \quad \sigma_i \mapsto \sigma_i y$$

then we can define isomorphism

$$M_{G_y}^G(\mathbb{Z}) = \mathrm{Hom}\ (\mathbb{Z}G, \mathbb{Z}) \to \bigoplus_{i=1}^r \mathbb{Z}y_i, \quad f \mapsto \sum_{i=1}^r f(\sigma_i) y_i \tag{9.2}$$

□

Example 9.22.

$$\begin{aligned}
\pi^{-1}(x) &= C_E \times_C \mathrm{Spec}\ k(x) = E \times_k C \times_C \mathrm{Spec}\ k(x) = E \times_k \mathrm{Spec}\ k(x) \\
&= \mathrm{Spec}\ (k(x) \otimes_k E) = \mathrm{Spec} \left(k(x) \otimes_k \frac{k[t]}{(f(t))} \right) \cong \mathrm{Spec}\ \frac{k(x)[t]}{f(t)} \\
&= \mathrm{Spec} \left(\prod_{i=1}^r k(x)[t]/(h_i(t)) \right) = \coprod_{i=1}^r \mathrm{Spec}\ \frac{k(x)[t]}{(h_i(t))}
\end{aligned}$$

Example 9.23. $k=\mathbb{Q}$, $E=\mathbb{Q}(i,\sqrt[4]{2})$, $k(x)\cong\mathbb{Q}(\sqrt{2})$

$$\text{Gal }(E/k)=\langle\sigma,\tau\mid\tau^2=e,\sigma^4=e,\tau\sigma\tau=\sigma^{-1}\rangle,\{\sigma(\sqrt[4]{2}=i\sqrt[4]{2})$$
$$\sigma(i)=i,\{\tau(\sqrt[4]{2}=\sqrt[4]{2})$$
$$\tau(i)=-i$$

$$E=\mathbb{Q}[x]/(x^8-4x^6+8x^4-4x^2+1)$$

$$k(x)\otimes_k E=\mathbb{Q}(\sqrt{2})[x]/(x^8-4x^6+8x^4-4x^2+1)=\frac{\mathbb{Q}(\sqrt{2})[x]}{(h_1(x))}\times\frac{\mathbb{Q}(\sqrt{2})[x]}{(h_2(x))}$$

Here $h_1(x)=x^4+(-\sqrt{2}-2)x^2-2\sqrt{2}+3$, $h_2(x)=x^4+(\sqrt{2}-2)x^2+2\sqrt{2}+3$. The stabilizer is $H=\langle e,\sigma^2,\tau,\sigma^2\tau\rangle$

Definition 9.24. The *residue map* $r_x^i(y)$ is defined to be

$$H^i(G,E(C_E)^\times)\to H^i(G,\text{Div }(C_E))\cong\bigoplus_{x\in C^{(1)}}H^i(G,M_{G_y}^G(\mathbb{Z}))\cong\bigoplus_{x\in C^{(1)}}H^i(G_y,\mathbb{Z})$$
$$\cong\bigoplus_{x\in C^{(1)}}H^{i-1}(G_y,\mathbb{Q}/\mathbb{Z})\to H^{i-1}(G_y,\mathbb{Q}/\mathbb{Z})$$

Lemma 9.25. For any $y,z\in\pi^{-1}(x)$, the residue maps $r_x^i(y),r_x^i(z)$ differ by a canonical isomorphism

Proof. There exists $\sigma\in G$ such that $\sigma y=z$, and $G_z=\sigma G_y\sigma^{-1}$

□

Lemma 9.26. The sequence

$$H^i(G,E(E_E)^\times)\xrightarrow{\bigoplus_{x\in C^{(1)}}r_x^i(y)}\bigoplus_{x\in C^{(1)}}H^{i-1}(G_y,\mathbb{Q}/\mathbb{Z})\xrightarrow{\sum_{x\in C^{(1)}}d_y\text{cor}_G^{G_y}}H^{i-1}(G,\mathbb{Q}/\mathbb{Z})$$

is a complex for $i\geq 2$, here $d_y=[E(y):E]$ is the degree of y

Proof. We can split (9.1) into two exact sequences

$$1\to E^\times\to E(C_E)^\times\to E(C_E)^\times/E^\times\to 1\tag{9.3}$$
$$1\to E(C_E)^\times/E^\times\to\text{Div }(C_E)\to\text{Pic }(C_E)\to 0\tag{9.4}$$

Hence we get commutative diagram with the help of Lemma 9.27

$$\begin{array}{ccccc}
H^i(G,E(C_E)^\times/E^\times) & \longrightarrow & H^i(G,\text{Div }(C_E)) & \longrightarrow & H^i(G,\text{Pic }(C_E)) \\
\uparrow & & \downarrow\cong & & \downarrow\cong \\
H^i(G,E(C_E))^\times & \longrightarrow & \bigoplus_{x\in C^{(1)}}H^i(G_y,\mathbb{Z}) & \longrightarrow & H^{i-1}(G,\mathbb{Q}/\mathbb{Z})\cong H^i(G,\mathbb{Z})
\end{array}$$

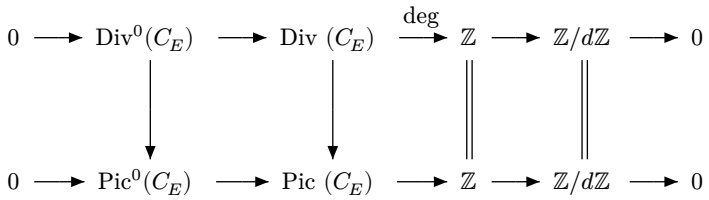
Looking at a single summand, the map reads

$$H^i(G,\text{Div }(C_E))\to H^i(G,M_{G_y}^G(\mathbb{Z}))\to H^i(G_y,\mathbb{Z})\to H^i(G,M_{G_y}^G(\mathbb{Z}))\xrightarrow{\phi_*}H^i(G,\mathbb{Z})$$

Recall $\phi:M_{G_y}^G(\mathbb{Z})=\text{Hom}_G(\mathbb{Z}[G],\mathbb{Z})\to\mathbb{Z}$ is defined via $\phi(f)=\sum_{s\in G/G_y}sf(s^{-1}e)$

□

Lemma 9.27. There is a commutative exact ladder



Here $\deg\left(\sum n_y y\right) = \sum n_y [E(y) : E]$
Proof. Hartshorne chapter II, corollary 6.10

□

Example 9.28. $C = \mathbb{P}^1_k$, $E(C_E) = E(t)^\times$, $G = \text{Gal } (E/k) = \text{Gal } (E(t)/k(t))$, then

$$H^2(G, E(C_E)^\times) \cong H^2(\text{Gal } (E(t)/k(t)), E(t)^\times) \cong \text{Br } (E(t)/k(t))$$

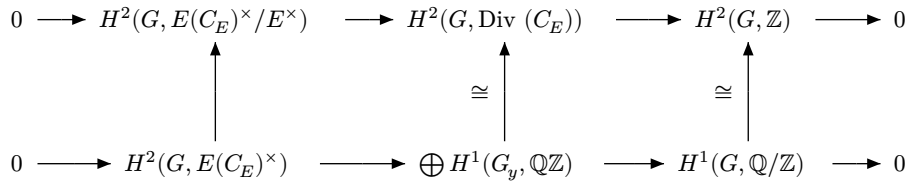
And our complex is

$$\text{Br } (E(t)/k(t)) \longrightarrow \bigoplus_{x \in (\mathbb{P}^1_k)^{(1)}} H^1(G_y, \mathbb{Q}/\mathbb{Z}) \longrightarrow H^1(G, \mathbb{Q}/\mathbb{Z})$$

Let $p(t) \in E(t)^\times$, assume E/k is cyclic with $\chi : \text{Gal } (E/k) \to \mathbb{Z}/n\mathbb{Z}$. Then $[(\chi, p(t))^{op}]$ is given by the 2 cocycle $\alpha : G \times G \to E(t)^\times$, $\alpha(\sigma^i, \sigma^j) = 1$ if $i + j < n$, $p(t)$ otherwise.

The residue map just takes div $(\alpha) = 0$, $[\infty] - [0]$
Theorem 9.29. Assume $C = \mathbb{P}^1_k$ such that $H^2(G, E^\times) = H^3(G, E^\times) = 0$ (This happens if k is a finite field), then 9.26 is exact
Proof. Pic $(\mathbb{P}^1_E) \underset{\cong}{\overset{\deg}{\to}} \mathbb{Z}$, then we get a splitting in (9.4) with $1 \mapsto [\infty]$.

so we get



□

Lemma 9.30. For any curve C/k as the in the set up

$$H^2(G, E(C_E)^\times) \cong H^2(\text{Gal } (E(C_E)/k(C)), E(C_E)^\times) \cong \text{Br } (E(C_E)/k(C))$$

Moreover, if k is perfect

$$\text{Br } (k(C)) = \varinjlim_{E/k \text{ finite Galois}} \text{Br } (E(C_E)/k(C)) = \varinjlim_{E/k \text{ finite Galois}} H^2(\text{Gal } (E(C_E)/k(C)), E(C_E)^\times)$$

Proof. The restriction

$$\text{Gal }(E(C_E)/k(C)) \to \text{Gal }(E/k), \varphi \mapsto \varphi\mid_E$$

is an isomorphism. This gives the first part. Suppose $k < E < F$ are all finite Galois, then we have inclusion

$$\text{Br }(E(C_E)/k(C)) \to \text{Br }(F(C_F)/k(C))$$

And all these map to Br $(k(C))$, surjective since any CSA/k, say A is split over $\bar{k}(C_{\bar{k}})$ by Tsen's theorem 8.8. This proves the moreover part

□

Example 9.31. Br $(\mathbb{F}_3(t)) \neq 0$, Take $k = \mathbb{F}_3$, $E = \mathbb{F}_{27}$, $C = \mathbb{P}^1_k$, we get split exact sequence

$$0 \longrightarrow \text{Br }(\mathbb{F}_{27}(t)/\mathbb{F}_3(t)) \longrightarrow \bigoplus H^1(G_y, \mathbb{Q}/\mathbb{Z}) \rightleftarrows H^1(G, \mathbb{Q}/\mathbb{Z}) \longrightarrow 0$$

So Br $(\mathbb{F}_{27}(t)/\mathbb{F}_3(t)) \cong \bigoplus_{x\in\mathbb{A}^1_k} H^1(G_y, \mathbb{Q}/\mathbb{Z}) = \bigoplus_{x\in\mathbb{A}^1_k, y^3-y-1 \text{ doesn't split in } k(x)} \mathbb{Z}/3\mathbb{Z}$

For every point $x \in \mathbb{A}^1_k$ with $k(x) = \mathbb{F}_3$, the local ring $\mathcal{O}_{\mathbb{A}^1,x}$ can be completed to $\widehat{\mathcal{O}_{\mathbb{A}^1,x}}$ with fraction field $\mathbb{F}_3((\lambda))$, where λ is a uniformizer of $\mathcal{O}_{\mathbb{A}^1,x}$, the map $O_{\mathbb{A}^1,x} \to \widehat{O_{\mathbb{A}^1,x}}$ induces $F(t) \to F((\lambda))$, giving Br $(\mathbb{F}_{27}(t)/\mathbb{F}_3(t)) \to$ Br $(\mathbb{F}_{27}((\lambda))/\mathbb{F}_3((\lambda))) \cong \mathbb{Z}/3\mathbb{Z}$